P101 · Unit 11 · Quick reference

Fields, potentials and accumulated work
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Recall **Field and response**	$F_{\mathrm{g}} = mg$; $F_{\mathrm{e}} = qE$. $[g] = \mathrm{N\,kg^{-1}}$; $[E] = \mathrm{N\,C^{-1}} = \mathrm{V\,m^{-1}}$. Signed q.
Recall **Radial sources**	Outward positive: $g_r = -GM / r^2$; $E_r = k_{\mathrm{e}} Q / r^2$. Point sources: $r > 0$. Spherically symmetric exterior: $r \geq R > 0$; fixed sources.
Recall **Accumulation**	$W = \int_a^b F_x(x)\,dx$ [J]; $J_x = \int_{t_a}^{t_b} F_{\mathrm{net},x}(t)\,dt = \Delta p_x$ [N s]. Start/end bounds; signed components; domain valid throughout.
Recall **Energy balance**	$\Delta K = W_{\mathrm{net}}$; $K = \frac{1}{2}mv^2$. Conservative field: $W_{\mathrm{field}} = -\Delta U$; $\Delta(K + U) = W_{\mathrm{other}}$.
Recall **Potential**	$U_{\mathrm{g}} = m\phi$; $\Delta U_{\mathrm{e}} = q\Delta V$. $[\phi] = \mathrm{J\,kg^{-1}}$; $[V] = \mathrm{J\,C^{-1}}$ (volt). $\Delta V = V_B - V_A$.
Recall **Point-source potential**	Zero at infinity: $\phi = -GM / r$; $V = k_{\mathrm{e}} Q / r$. Potentials add as scalars; fields as vectors. Exterior/point-source domains as above.

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Understand **Work from endpoints**	$W_g = GMm(1 / b - 1 / a)$. $W_e = k_e Qq(1 / a - 1 / b) = -q\Delta V$. Radial paths; fixed sources; same domain throughout.
Understand **Slope and reference**	$E_x = -dV / dx$; $g_r = -d\phi / dr$. Uniform E_x: $\Delta V = -E_x \Delta x$. Constant reference shifts leave fields and differences unchanged.
Understand **Magnetic response**	Compass north end aligns with B [tesla, T]. Classical point charge: magnetic force is zero at rest or parallel motion; otherwise perpendicular to velocity and field. Magnetic force does no work on that particle.
Consult **Constants and scales**	$G \approx 6.67\times10^{-11}\ \mathrm{N\,m^2\,kg^{-2}}$. $k_e \approx 8.99\times10^{9}\ \mathrm{N\,m^2\,C^{-2}}$. $e \approx 1.602\times10^{-19}$ C; nC: 10^{-9} C; µC: 10^{-6} C.
Understand **Numerical work**	Midpoint: $h = (b - a) / n$, $r_i = a + (i + \frac{1}{2})h$. $W_n = \sum_{i=0}^{n-1} (C / r_i^2)h$. Benchmark: $C(1 / a - 1 / b)$. Check domain, units, hand case, reversal, scaling and refinement.

Consult · Python

```
# Full guarded functions: Lesson 4 / notebook
C = -GM * m           # gravity; SI units
C = k_e * Q * q       # electric alternative
estimate = radial_work(C, a, b, n, radius)
exact = exact_work(C, a, b, radius)
error_J = abs(estimate - exact)
# Increase n; compare errors, then reverse a,b.
# Require positive exterior radii and integer n > 0.
```

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